

Assignment 1 DESIGN.pdf:

Description of Program:

This program simulates a simplified version of David Moffat's dice game Pass the Pigs. It involves using all the basic programming language facilities: loops, conditions, arrays, primitive types, etc. The game can be played with a minimum of 2 players and a maximum of 10 players. The players are arranged in a circle and take turns rolling a die (the die is called Pig) to earn points. Rolling Side earns the player 0 points and his turn is officially over. The Pig then gets passed to the next player. Rolling either Razorback or Trotter earns 10 points for the player. Rolling Snouter earns 15 points. Lastly, rolling Jowler earns 5 points. The game ends when any player has earned 100 or more points.

Files to be included in directory "asgn1":

- Pig.c
 - Source file that contains main() and the majority of the content that allows the game to simulate.
- names.h
 - This header file contains the array of player names to be used in the implementation of the game.
- Makefile
 - The following Makefile targets are implemented and function properly:
 - make
 - make all
 - make clean
 - make format
 - The Makefile also must build the program correctly.

- README.md
 - in Markdown
 - briefly describes program
 - describes how to build program
 - describes how to run program
- DESIGN.pdf
 - Describes design for the program thoroughly with pseudocode

Pseudocode/Structure:

- Get user input for the number of players (must be valid integer between 2 and 10 inclusive)
 - If invalid user input then print error message "Invalid number of players.
Using 2 instead .\n"
 - Use default value of 2 for the number of players
- Get user input for seed
 - If invalid user input then print error message "Invalid random seed.
Using 2021 instead .\n"
 - Use default value 2021 for seed
- Set random seed so that each player starts off with 0 points
- Go in a circle starting at player 0
 - Print player name (index 0 from array names.h is player 0, index 1 is player 1, etc.)
 - Player rolls pig
 - If player doesn't roll on one of its side, increment point total until 100 points or player rolls on one of its sides
 - If player rolls on one of its sides, go to next player

- If player has greater or equal to 100 points, print congratulatory message indicating player has won the game